

CUTANEOUS

MANIFESTATIONS OF DRUG ABUSE AT A GLANCE

Parenteral administration of drugs can lead to skin lesions and eruptions through local or systemic effects, including toxic or hypersensitivity reactions caused by the drug, adulterants, or infectious agents. Intravenous drug use often produces characteristic skin findings that may indicate illicit drug use to those familiar with such signs.

DEFINITIONS

- **Parenteral drug use:** Administration of drugs via routes other than the gastrointestinal tract, most commonly intravenous, intramuscular, or subcutaneous injection.
- **Cutaneous manifestations:** Skin changes including tracks, scars, ulcers, infections, and inflammatory reactions associated with drug injection or systemic effects of abused substances.

COMMONLY ABUSED DRUGS

- **Cannabis** (marijuana, pot, weed, ganja): Derived from *Cannabis sativa*, it contains delta-9-tetrahydrocannabinol (THC), the primary psychoactive compound.
- **Diacetylmorphine** (heroin): A potent opioid that accounts for 90% of opiate abuse in the United States. When injected intravenously, it produces a "rush" within 7–8 seconds; when snorted or smoked, the effect occurs in 10–15 minutes.
- **Cocaine** (coke, blow, snow): A stimulant and topical anesthetic derived from the coca plant. It is commonly used nasally, intravenously, or smoked.

- **Crack cocaine:** The freebase form of cocaine, smoked for rapid and intense euphoria. It is highly addictive.
- **Methamphetamine** (meth, crystal meth, ice): A powerful central nervous system stimulant. When crystallized ("ice"), it is typically inhaled or injected to produce a strong rush.

CUTANEOUS SIGNS OF INJECTABLE DRUG ABUSE

The pattern and appearance of skin lesions can help identify current or past drug dependency. The most specific cutaneous indicators include:

- **Track marks:** Linear hyperpigmented scars or indurations along veins, especially in the antecubital fossa, forearm, or dorsum of the hand.
- **Puncture wounds:** Small scabs or crusted papules at injection sites.
- **Ecchymoses:** Bruising around injection areas.
- **Phlebitis:** Inflammation of veins, presenting as tender, erythematous cords.
- **Woody dermal induration:** Firm, fibrotic skin changes due to repeated injections and sclerosing phlebitis.
- **Cutaneous ulcers:** Often located in atypical or hidden areas (e.g., groin, feet, neck), resulting from poor injection technique, contaminated substances, or vasculitic effects.
- **Scars of unknown origin:** Multiple, irregular scars suggestive of self-injection practices.

INFECTIOUS COMPLICATIONS

Drug addicts are at increased risk for:

- **Blood-borne infections:**
 - Human immunodeficiency virus (HIV): Approximately 40% of AIDS-related deaths are linked to drug abuse.
 - Hepatitis B and C: About 60% of new hepatitis C cases in the U.S. result from needle sharing.
 - One-third of injection drug users become infected

with hepatitis C within 6 months of starting injection. - By 2 years, up to 90% may be infected. - Hepatitis D (requires HBV co-infection) - Human T-lymphotropic virus types 1 and 2 (HTLV-1/2)

- **Bacterial infections:** - Predominantly affect skin and soft tissue (e.g., abscesses, cellulitis, necrotizing fasciitis). - Often polymicrobial or caused by antibiotic-resistant organisms (e.g., MRSA). - Culture-guided antibiotic therapy is essential due to high rates of unusual pathogens and resistance.

ASSOCIATED HEALTH AND SOCIAL PROBLEMS

- Increased prevalence of sexually transmitted infections (e.g., syphilis, gonorrhea).
- High rates of depression, anxiety, and other psychiatric disorders.
- Accidental injuries, trauma, and overdose.
- Exposure to criminal violence and domestic abuse.
- Societal burdens including:
 - Loss of work productivity
 - Increased crime and incarceration
 - Property damage
- Economic cost: \$143.4 billion in 1998, projected to rise to \$204 billion by 2007 (9% attributed to health care costs).

CLINICAL RECOGNITION

The typical appearance of chronic stimulant users ("speed freaks") includes: - Pale, sagging, sweaty complexion - Poor hygiene and body odor - Cachexia - Dental decay ("meth mouth") - Skin ulcers and scarring from picking or injection

Recognition of these cutaneous and systemic signs can aid in early intervention, treatment, and prevention of further complications.